

Project RoyalDev UseCases:

1.

Title	UC-AUTH-01: Register an account
Actors	ACT-01 (Registered user)
Preconditions	User has navigated to the web application URL and is not authenticated.
Basic Flow	1. Prospective user accesses the registration page via the application frontend. 2. User provides required details (e.g., email address, full name, preferred password). 3. Backend validates input formats, checks for existing user conflicts, and applies a secure password hashing function. 4. Backend creates the user record in the relational database. 5. User logs in with established credentials; the system authenticates the user and routes them to the main Dashboard.
Alternate Flow	Invalid / Duplicate Registration Details: System flags invalid formatting or existing email addresses, rendering user-friendly validation messages without exposing system architecture details. Invalid Credentials: System alerts the user of incorrect email/password combinations while maintaining rate limiting to prevent brute-force attacks.
Postconditions	User account is created and stored in database; user is authenticated and redirected to Dashboard.
Open Issues	

2.

Title	UC-AUTH-02: Sign in
Actors	ACT-01 (Registered user)
Preconditions	User is at the sign in page with registered credentials.

Basic Flow	6. User enters registered credentials. 7. System validates credentials against the database. 8. User is granted access and directed to the Dashboard.
Alternate Flow	Invalid Credentials: System alerts user of bad credentials and rate-limits login attempts.
Postconditions	User session established and user redirected to Dashboard.
Open Issues	

3.

Title	UC-AUTH-03: Persistent User Sessions / Authentication State Management
Actors	ACT-01 (Registered user)
Preconditions	User has previously completed authentication and holds a valid refresh token or secure cookie session.
Basic Flow	1. User reopens or reloads the web application in a browser. 2. System checks secure client storage for active session tokens. 3. Backend validates session tokens and returns active user state seamlessly without requesting credentials. 4. User is restored to their authenticated dashboard view.
Alternate Flow	Expired / Invalid Session: System clears invalid session tokens and redirects user to the login screen with a session timeout notification.
Postconditions	Active session token is validated and user access is restored to authenticated state.
Open Issues	

4.

Title	UC-BUD-01: Create a monthly budget
Actors	ACT-01 (Registered user)

Preconditions	User is successfully authenticated and present on the Budget Management interface.
Basic Flow	1. User navigates to the Budget section of the dashboard. 2. User selects a target month and defines budget allocations across specific financial categories (e.g., Housing, Groceries, Utilities, Entertainment). 3. System validates that all entered allocation amounts are non-negative numeric values. 4. System updates the relational database and recalculates real-time category progress bars on the user interface.
Alternate Flow	Negative or Non-Numeric Input: System highlights input fields with an error notification and prevents form submission until corrected.
Postconditions	Monthly category budget allocations are saved to database and displayed with active progress
Open Issues	

5.

Title	UC-FIN-01: Log simulated transaction
Actors	ACT-01 (Registered user)
Preconditions	User is authenticated and possesses active budget allocations for the current billing cycle.
Basic Flow	1. User submits a transaction entry or initiates a simulated bill payment using dummy financial card inputs. 2. System validates input fields against predefined dummy card formats and rules. 3. System writes the transaction log to the database and updates category progress against total budget limits.
Alternate Flow	Malformed Card Entry: System alerts user of bad test card formats and requests input adhering to dummy testing patterns.
Postconditions	Transaction log is written to the database and category budget limits are recalculated
Open Issues	

6.

Title	UC-AI-01: Generate AI budget insights
Actors	ACT-01 (Registered user), ACT-02 (AI/LLM Service)
Preconditions	User has created at least one category budget and logged relevant transaction data.
Basic Flow	1. User accesses the assistant chat interface. 2. Backend compiles anonymized spending and budget data into a structured system prompt for the AI endpoint. 3. ACT-02 (AI/LLM Service) processes the context and returns actionable, customized recommendations. 4. Frontend displays formatted insights cleanly on the user interface.
Alternate Flow	AI Service Timeout/Unavailable: System captures API communication failures gracefully and presents a fallback message advising the user to try again shortly.
Postconditions	Formatted AI recommendations are displayed cleanly on user interface
Open Issues	

7.

Title	UC-ADM-01: Monitor operations & accounts
Actors	System administrator
Preconditions	Administrator is authenticated with administrative credentials.
Basic Flow	9. Administrator accesses admin console. 10. System displays operational statistics and account oversight options. 11. Administrator reviews system metrics and manages active accounts as needed.
Alternate Flow	Unauthorized Access Attempt: Non-admin users are denied access and redirected to the standard user dashboard.
Postconditions	Administrative operations and user account states are updated or audited in the system logs
Open Issues	

8.

Title	UC-PROF-01: Display Profile Info and Settings
Actors	ACT-01 (Registered user)
Preconditions	User is authenticated and logged into the application.
Basic Flow	1. User navigates to the Profile & Settings page. 2. System retrieves user details (e.g., name, email, notification preferences) and displays them on the interface. 3. User updates preference settings or profile information and submits the changes. 4. Backend validates the modifications and persists updated records to the database.
Alternate Flow	Invalid Input: System highlights invalid fields and displays relevant error messages preventing submission.
Postconditions	User profile and setting modifications are validated and stored in the database.
Open Issues	